

No monomial statistic on cylindric tableaux carries Warnaar’s generalised affine Jacobi–Trudi determinant

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10 October 2026 (PROVE 2026-10-10)

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Abstract

Warnaar [arXiv:2511.17034, §6] asks whether Huh–Kim–Krattenthaler–Okada’s combinatorial evaluation of a $t = 1$ affine bounded Littlewood determinant [arXiv:2301.13117, Thm. 3.3] extends to his generalised determinant (1) “*by introducing an additional statistic on cylindric tableaux*”. We prove that it does not, if “statistic” is read in the usual sense of a $\mathbb{Z}$ -valued function $T \mapsto \text{stat}(T)$ contributing a single power $t^{\text{stat}(T)}$. The obstruction is a cardinality count, not a sign or a symmetry. For every $\ell \geq 0$, setting $M = \ell + 2$ and $k = 1$, $n = 2M$, the coefficient of $x_1 \cdots x_n$ in Warnaar’s determinant is $C_M - (2M - 1)t^{M-1} + t^{M+1}$ with C_M a Catalan number, while the cylindric tableaux available to carry it number C_M with one sign and only $2M - 2$ with the other: the coefficient $-(2M - 1)$ asks for one object more than exists. At $\ell = 0$ the failure is stronger still — the coefficient is $2 - 3t + t^3$, whose coefficients have absolute-value sum 6, while the total number of $(2, 2)$ -cylindric tableaux of that content, over *all* shapes and not only the two Huh–Kim–Krattenthaler–Okada use, is 4 — so not even re-choosing the signs helps. Consequently no twisted transfer-matrix trace $\text{Tr}(DT^N)$ with a diagonal twist D can produce the determinant either: a twist contributes one monomial per configuration, and the target needs more than one monomial’s worth per configuration. For $\ell \geq 1$ the obstruction weakens in a way we make precise: the strict form of the question (Thm. 3.3 with $t^{\text{stat}(T)}$ inserted) still fails, but the slack created by the cylindric shapes that Thm. 3.3 assigns weight 0 is enough to leave the re-signed form open. Since no such slack exists at $\ell = 0$, any uniform answer must attach a *polynomial* in t to each cylindric tableau — which is exactly the shape of Macdonald’s classical $\psi_T(t)$.

1 Provenance

The question addressed here is Warnaar’s, quoted verbatim in §2. The idea of looking for it on the integrable side — as a diagonal twist of a cylindric transfer matrix — came to me from a conference talk; that talk is *not* an input to anything below and no construction here is attributed to it. Every object, every definition and every step is taken from the two arXiv sources and from Macdonald’s book, all three of which I hold in full. Neither Warnaar’s paper nor Huh et al.’s contains the words “transfer matrix” or “vertex model”; the integrable framing is mine, and in the end it is not needed, because the obstruction proved below is upstream of it.

2 Setup and the two verbatim inputs

Fix integers $k \geq 1$, $\ell \geq 0$, $n \geq 1$ and set

$$w := 2\ell + 2, \quad M := k + \ell + 1, \quad N := 2M = 2k + w.$$

Write $x = (x_1, \dots, x_n)$, let $e_r = e_r(x)$ be the elementary symmetric functions ($e_r = 0$ for $r < 0$ or $r > n$), and let $\dot{e}_r := e_r(x_1, x_1^{-1}, \dots, x_n, x_n^{-1})$. Following Huh–Kim–Krattenthaler–Okada (“HKKO” below) put

$$f_r(x) := \sum_{i \in \mathbb{Z}} e_i(x) e_{i+r}(x), \quad F_{r,N}(x) := \sum_{y \in \mathbb{Z}} f_{r+Ny}(x),$$

so that $f_r = f_{-r}$ and $F_{r,N} = F_{-r,N}$.

Warnaar’s object

Warnaar [3, §6, eq. (Eq-generalise)] asks for the expansions of

$$\mathcal{G}_{k,\ell,n}(x;t) := \sum_{y \in \mathbb{Z}^k} \det_{1 \leq i,j \leq k} \left(t^{My_i^2 - jy_i} (\dot{e}_{n-i+j-Ny_i}(x) - \dot{e}_{n+i+j-Ny_i}(x)) \right). \quad (1)$$

He then writes, verbatim:

Recalling the notation (Eq-FiN), (Eq-generalise) for $t = 1$ is

$$(x_1 \cdots x_n)^{-k} \det_{1 \leq i,j \leq k} (F_{i-j,2(k+\ell+1)}(x) - F_{i+j,2(k+\ell+1)}(x)),$$

for which Huh et al. [HKKO25, Theorem 3.3] obtained a combinatorial expression in terms of cylindric tableaux as follows:

$$(x_1 \cdots x_n)^{-k} \left(\sum'_{\lambda \in \text{Par}_{n,2k}^{2\ell+2}} - \sum''_{\lambda \in \text{Par}_{n,2k}^{2\ell+2}} \right) \sum_{T \in \text{CSSYT}_{n;2k,2\ell+2}(\lambda)} x^T.$$

*Here the prime in the first sum denotes the restriction that λ must be even and the double prime in the second sum denotes the restrictions that $\lambda'_1 - \lambda'_{2k} = 2\ell + 2$, the last $2\ell + 2$ parts of λ are odd and the other parts are even. **Is it possible to extend this result to (Eq-generalise) by introducing an additional statistic on cylindric tableaux?***

(Boldface mine. The question is the last sentence.)

HKKO Theorem 3.3

HKKO [1, Thm. 3.3], in their own notation. For positive integers h, w let $\text{Par}(h, w)$ be the set of partitions λ with $\ell(\lambda) \leq h$ and $\lambda_1 - \lambda_h \leq w$. For λ with $\ell(\lambda) \leq 2h$ and $\lambda_1 - \lambda_{2h} \leq w$ set

$$c_{2h,w}^\pm(\lambda) = \begin{cases} 1, & \text{if } \lambda_{2i-1} = \lambda_{2i} \text{ for all } 1 \leq i \leq h, \\ \pm 1, & \text{if } \lambda_1 - \lambda_{2h} = w \text{ and } \lambda_{2i} = \lambda_{2i+1} \text{ for all } 1 \leq i \leq h-1, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

Then (their equation (3.6))

$$\sum_{\lambda \in \text{Par}(2h, w)} c_{2h, w}^-(\lambda) s_{\lambda[2h, w]'}(x) = \det_{1 \leq i, j \leq h} (F_{-i+j, 2h+w}(x) - F_{i+j, 2h+w}(x)), \quad (3)$$

and, by [1, Prop. 2.9], $s_{\lambda[2h, w]'}(x) = \sum_{T \in \text{CRST}(\lambda; 2h, w)} x^T$, the sum being over $(2h, w)$ -cylindric row-strict tableaux of shape λ . A tableau is (h, w) -cylindric when it has at most h rows and $T \cup (T + (h, -w))$ is again a tableau of a valid skew shape; in particular $\text{CRST}(\lambda; 2h, w) = \emptyset$ unless $\lambda \in \text{Par}(2h, w)$, so (3) already ranges over *all* cylindric tableaux of this type. Throughout, $h = k$ and $N = 2h + w = 2k + 2\ell + 2$, which is Warnaar's $2(k + \ell + 1)$.

We also use HKKO's lattice-path form [1, Prop. 2.7]: the number of $T \in \text{CRST}(\lambda; m, w)$ with content α equals the number of paths in $\mathbb{Z}^m$ from the origin to $(\lambda_1, \dots, \lambda_m)$ staying in $\{x_1 \geq x_2 \geq \dots \geq x_m \geq x_1 - w\}$, the i -th step being a 0/1 vector with α_i ones.

3 The gate, and that it is a gate

Lemma 1. $e_{n-r}(x) = (x_1 \cdots x_n)^{-1} f_r(x)$ for all $r \in \mathbb{Z}$, and consequently

$$(x_1 \cdots x_n)^k \mathcal{G}_{k, \ell, n}(x; 1) = \det_{1 \leq i, j \leq k} (F_{-i+j, N} - F_{i+j, N}) = \sum_{\lambda \in \text{Par}(2k, w)} c_{2k, w}^-(\lambda) \sum_{T \in \text{CRST}(\lambda; 2k, w)} x^T.$$

Proof. The first identity is Warnaar's [3, proof of Lem. 5.2]; it is also immediate from $e_r(x^\pm) = \sum_m e_m(x) e_{m+r-n}(x) (x_1 \cdots x_n)^{-1}$. For the second, the (i, j) entry of the matrix in (1) at $t = 1$ is, by the first identity, $(x_1 \cdots x_n)^{-1} (f_{(i-j)+Ny_i} - f_{(i+j)-Ny_i})$, in which the dependence on y is through y_i alone. Hence the sum over $y \in \mathbb{Z}^k$ may be carried into row i by multilinearity of the determinant in its rows, and $\sum_{y \in \mathbb{Z}} f_{(i-j)+Ny} = F_{i-j, N}$, $\sum_{y \in \mathbb{Z}} f_{(i+j)-Ny} = F_{i+j, N}$. Since $F_{r, N} = F_{-r, N}$ this is the middle expression. The right-hand equality is (3) with $h = k$. $\square$

Remark 2 (where the difficulty is). At $t = 1$ the weight $t^{My_i^2 - jy_i}$ is absent and the y -sum passes into a single row. For $t \neq 1$ that weight depends on the *column* index j as well as on y_i , so it is not a row constant and the y -sum cannot be carried in. This is precisely what Warnaar's question asks one to repair.

Lemma 1 was verified symbolically (exact arithmetic, SymPy) at $(k, \ell, n) = (1, 0, 4), (1, 0, 5), (1, 1, 5), (1, 1, 6), (1, 1, 7)$ and (3) itself at $(h, w, n) = (1, 1, 4), (1, 2, 3), (1, 2, 4), (1, 4, 4), (1, 4, 5), (2, 2, 4)$.

The gate discriminates. Three deliberately wrong weight functions in place of $c_{2h, w}^-$ — namely $c_{2h, w}^+$, the unsigned version, and the version with the wall condition $\lambda_1 - \lambda_{2h} = w$ replaced by $w - 1$ — all *fail* (3) at each of $(h, w, n) = (1, 2, 4), (1, 4, 5), (2, 2, 4)$, while the true c^- passes. So the $t = 1$ comparison is not a constant function of the claim.

The transcription of (1) is confirmed by a second mechanism. At $\ell = 0$, (1) is Warnaar's Conjecture 1.5 (his (Eq.PCn)), which asserts $\mathcal{G}_{k, 0, n}(x; t) = (x_1 \cdots x_n)^{-k} \sum_{\lambda \text{ even}, \lambda_1 \leq 2k} P_\lambda(x; t)$. Computing the right-hand side from Macdonald's symmetrisation formula for Hall–Littlewood polynomials — an engine sharing no code with (1) — confirms the identity exactly at $(n, k) = (3, 1), (4, 1), (3, 2)$. Six perturbations of the exponent $My_i^2 - jy_i$ were tested against this gate:

replacing $-jy_i$ by $+jy_i$, deleting it, and changing M to $M \pm 1$ or My_i^2 to $|y_i|$ all fail. One arm is **vacuous** and is recorded as such: replacing $-jy_i$ by $-iy_i$ (column index by row index) *passes* at $k = 1$ and $k = 2$, so this gate cannot see which index carries the linear term. Nothing below depends on that distinction.

4 The obstruction

Definition 3. Fix (k, ℓ, n) and let $\mathcal{T}$ denote the set of all $(2k, w)$ -cylindric row-strict tableaux with entries in $\{1, \dots, n\}$, of all shapes $\lambda \in \text{Par}(2k, w)$. For $T \in \mathcal{T}$ let $\lambda(T)$ be its shape.

(S) Warnaar's question has a *monomial solution* at (k, ℓ, n) if there is $\text{stat} : \mathcal{T} \rightarrow \mathbb{Z}$ with

$$(x_1 \cdots x_n)^k \mathcal{G}_{k, \ell, n}(x; t) = \sum_{T \in \mathcal{T}} c_{2k, w}^-(\lambda(T)) t^{\text{stat}(T)} x^T.$$

This is literally (3) with $t^{\text{stat}(T)}$ inserted, i.e. the display Warnaar quotes, graded.

(S $^\pm$) It has a *signed monomial solution* if there are $\varepsilon : \mathcal{T} \rightarrow \{0, \pm 1\}$ and $\text{stat} : \mathcal{T} \rightarrow \mathbb{Z}$ with

$$(x_1 \cdots x_n)^k \mathcal{G}_{k, \ell, n}(x; t) = \sum_{T \in \mathcal{T}} \varepsilon(T) t^{\text{stat}(T)} x^T.$$

Clearly (S) $\Rightarrow$ (S $^\pm$). Both put a single power of t on each cylindric tableau; (S $^\pm$) additionally allows the sign to be re-chosen tableau by tableau and allows the shapes with $c^- = 0$ to be recruited.

For a content $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$ write

$$c_\alpha(t) := [x^\alpha] (x_1 \cdots x_n)^k \mathcal{G}_{k, \ell, n}(x; t) \in \mathbb{Z}[t, t^{-1}], \quad \|c_\alpha\|_1 := \sum_e |[t^e]c_\alpha|,$$

and let

$$A_\alpha := \#\{T \in \mathcal{T} : \text{cont}(T) = \alpha, c^-(\lambda(T)) = +1\}, \quad B_\alpha := \#\{\cdots = -1\}, \quad \text{TOT}_\alpha := \#\{T \in \mathcal{T} : \text{cont}(T) = \alpha\}.$$

By Lemma 1, $c_\alpha(1) = A_\alpha - B_\alpha$ for every α .

Lemma 4 (capacity). *Fix (k, ℓ, n) and a content α .*

1. *If (S) holds then $\sum_e \max([t^e]c_\alpha, 0) \leq A_\alpha$ and $\sum_e \max(-[t^e]c_\alpha, 0) \leq B_\alpha$.*
2. *If (S $^\pm$) holds then $\|c_\alpha\|_1 \leq \text{TOT}_\alpha$.*

Proof. (1) Let a_e (resp. b_e) be the number of $T \in \mathcal{T}$ of content α with $c^-(\lambda(T)) = +1$ (resp. -1) and $\text{stat}(T) = e$. These are nonnegative integers with $\sum_e a_e = A_\alpha$, $\sum_e b_e = B_\alpha$, and comparing the coefficient of $x^\alpha t^e$ gives $[t^e]c_\alpha = a_e - b_e$. Hence $\sum_e \max(a_e - b_e, 0) \leq \sum_e a_e = A_\alpha$, and symmetrically. (2) Likewise $[t^e]c_\alpha = \sum_T \varepsilon(T) [\text{stat}(T) = e]$ over the T of content α , so $\sum_e |[t^e]c_\alpha| \leq \text{TOT}_\alpha$. $\square$

The content of Lemma 4 is that a monomial weight can only *redistribute* objects among powers of t ; it cannot create them. The ℓ^1 -norm of the answer is therefore bounded by the number of objects available to carry it. We now exhibit a content where it is not.

Theorem 5 (main). *Let $\ell \geq 0$, put $M := \ell + 2$ and take*

$$k = 1, \quad w = 2\ell + 2 = 2M - 2, \quad N = 2M, \quad n = N = 2M, \quad \alpha = (1^n).$$

Then, with $C_M = \frac{1}{M+1} \binom{2M}{M}$ the M -th Catalan number,

$$c_\alpha(t) = C_M - (2M - 1)t^{M-1} + t^{M+1}, \quad A_\alpha = C_M, \quad B_\alpha = 2M - 2. \quad (4)$$

Consequently (S) fails for every $\ell \geq 0$: Huh–Kim–Krattenthaler–Okada’s Theorem 3.3 with $t^{\text{stat}(T)}$ inserted is false for every statistic stat , at every ℓ .

Moreover, for $\ell = 0$ (so $M = 2$, $n = 4$) one has $\text{TOT}_\alpha = A_\alpha + B_\alpha = 4$ while $\|c_\alpha\|_1 = 6$, so $(S^\pm)$ fails as well: no signed monomial statistic on cylindric tableaux exists for Warnaar’s Conjecture (EqPCn).

Proof. Step 1: the coefficient. For $k = 1$ the determinant in (1) is 1×1 with $i = j = 1$, so by Lemma 1’s first identity

$$(x_1 \cdots x_n) \mathcal{G}_{1,\ell,n}(x; t) = \sum_{y \in \mathbb{Z}} t^{My^2 - y} (f_{Ny}(x) - f_{Ny-2}(x)),$$

where we used $f_{-r} = f_r$ to rewrite $\dot{e}_{n+2-Ny} = (x_1 \cdots x_n)^{-1} f_{Ny-2}$. Since $n = N$ and $f_r = 0$ for $|r| > n$, the term f_{Ny} vanishes unless $|y| \leq 1$, and f_{Ny-2} vanishes unless $y \in \{0, 1\}$ (for $y = -1$ one has $|-N-2| = N+2 > n$; for $|y| \geq 2$, $|Ny-2| \geq 2N-2 > N = n$ because $N \geq 4$). The surviving exponents are $My^2 - y = 0, M-1, M+1$ at $y = 0, 1, -1$, so

$$(x_1 \cdots x_n) \mathcal{G}_{1,\ell,n} = (f_0 - f_2) + t^{M-1}(f_N - f_{N-2}) + t^{M+1}f_N. \quad (5)$$

Now extract $[x_1 x_2 \cdots x_n]$. For $a+b = n$ one has $[x_1 \cdots x_n] e_a e_b = \binom{n}{a}$, since the monomial is square-free of degree n and the two factors must use complementary variable sets; and $[x_1 \cdots x_n] e_a e_b = 0$ when $a+b \neq n$. Hence in $f_r = \sum_m e_m e_{m+r}$ exactly one term survives, the one with $2m+r = n$:

$$[x_1 \cdots x_n] f_0 = \binom{2M}{M}, \quad [x_1 \cdots x_n] f_2 = \binom{2M}{M-1}, \quad [x_1 \cdots x_n] f_{N-2} = \binom{n}{1} = 2M, \quad [x_1 \cdots x_n] f_N = \binom{n}{0} = 1.$$

(For f_2 : $2m+2 = n = 2M$ gives $m = M-1$. For $f_{N-2} = f_{n-2}$: $m = 1$. For $f_N = f_n$: $m = 0$.) Substituting into (5) and using the standard identity $\binom{2M}{M} - \binom{2M}{M-1} = C_M$ gives

$$c_\alpha(t) = C_M + t^{M-1}(1 - 2M) + t^{M+1},$$

which is (4).

Step 2: the counts. Content $\alpha = (1^n)$ forces $|\lambda| = n$, and by HKKO’s lattice-path description each of the n steps is $(1, 0)$ or $(0, 1)$, so the difference $d = x_1 - x_2$ performs a ± 1 walk of length $n = 2M$ from $d = 0$, confined to $[0, w] = [0, 2M-2]$ by the region condition $x_1 \geq x_2 \geq x_1 - w$.

For A_α the endpoint is $d = 0$ (shape (M, M) , the unique even shape of size n with $\lambda_1 = \lambda_2$). A nonnegative ± 1 path of length $2M$ returning to 0 is a Dyck path, and there are C_M of them; its height is at most $M \leq 2M-2$ for $M \geq 2$, so the ceiling is never active and no path is lost. Hence $A_\alpha = C_M$.

For B_α the endpoint is $d = w = 2M-2$ (shape $(2M-1, 1)$). With u up-steps and δ down-steps, $u + \delta = 2M$ and $u - \delta = 2M-2$, so $\delta = 1$: the walk is all up except for a single down-step, say

at position p . Its value after p steps is $p - 2$, which must be ≥ 0 , so $p \geq 2$; and its maximum is $\max(p - 1, 2M - 2)$, which must be $\leq 2M - 2$, so $p \leq 2M - 1$. Both conditions are also sufficient. Hence $p \in \{2, \dots, 2M - 1\}$ and $B_\alpha = 2M - 2$.

Step 3: (S) fails. Suppose (S) held. In the notation of Lemma 4(1), comparing coefficients of t^{M-1} gives $a_{M-1} - b_{M-1} = -(2M - 1)$, so $b_{M-1} \geq 2M - 1$. But $\sum_e b_e = B_\alpha = 2M - 2 < 2M - 1$, a contradiction. This holds for every $M \geq 2$, i.e. every $\ell \geq 0$.

Step 4: (S $^\pm$) fails at $\ell = 0$. Here $M = 2$, $n = 4$, $w = 2$, and (4) reads $c_\alpha(t) = 2 - 3t + t^3$ with $A_\alpha = B_\alpha = 2$. The shapes of size 4 in $\text{Par}(2, 2)$ are $(2, 2)$ and $(3, 1)$, since $(4, 0)$ violates $\lambda_1 - \lambda_2 \leq w = 2$; these are exactly the shapes carrying $c^- = +1$ and $c^- = -1$. Hence $\text{TOT}_\alpha = A_\alpha + B_\alpha = 4$, while $\|c_\alpha\|_1 = 2 + 3 + 1 = 6 > 4$, and Lemma 4(2) applies. $\square$

Remark 6 (why $\ell = 0$ is the only place the stronger form bites). The deficit in Step 3 is $(2M - 1) - (2M - 2) = 1$, uniformly in ℓ — the strict form fails by exactly one tableau, always. The weaker form (S $^\pm$) is a different matter: it needs $\|c_\alpha\|_1 = C_M + 2M$ to exceed TOT_α , the number of *all* confined walks of length $2M$ in $[0, 2M - 2]$. For $M = 2$ the only reachable endpoints are $d = 0$ and $d = 2 = w$, both of which HKKO's theorem already uses, so $\text{TOT} = A + B = 4 < 6$. For $M \geq 3$ the intermediate even endpoints $d = 2, 4, \dots, w - 2$ become available; these are shapes with $c_{2,w}^-(\lambda) = 0$, invisible to Theorem 3.3, and they supply enough slack that $\|c_\alpha\|_1 < \text{TOT}_\alpha$ — at $M = 3$, $\text{TOT} = 18$ against $\|c_\alpha\|_1 = 11$. So the escape route available for $\ell \geq 1$ is precisely “recruit the shapes the theorem assigns weight zero”, and at $\ell = 0$ there are none to recruit in even degree.

Theorem 7 (the obstruction is not confined to $k = 1$). *At $(k, \ell, n) = (2, 0, 4)$ and content $\alpha = (2, 2, 2, 2)$,*

$$c_\alpha(t) = 10 - 16t + 11t^2 - 4t^3, \quad A_\alpha = 6, \quad B_\alpha = 5, \quad \text{TOT}_\alpha = 11, \quad \|c_\alpha\|_1 = 41.$$

Hence (S $^\pm$) fails, by a margin of 30. Across all 313 monomials at these parameters, 145 violate (S $^\pm$).

Scope of the computation. Theorem 5 was found by, and is consistent with, an exhaustive sweep. For each listed (k, ℓ, n) every monomial of $(x_1 \cdots x_n)^k \mathcal{G}$ was examined; the identity $c_\alpha(1) = A_\alpha - B_\alpha$ was asserted for *every* one of them (and holds), and the two capacity inequalities were tested. See Table 1. The counts $A_\alpha, B_\alpha, \text{TOT}_\alpha$ were computed twice by independent mechanisms — from the cylindric Jacobi–Trudi determinant [1, Thm. 2.10] and from the lattice-path model [1, Prop. 2.7] — agreeing on all 476 contents compared at $(k, w, n) = (1, 2, 4), (1, 4, 5), (2, 2, 4)$.

5 What this says about the twisted-trace ansatz

The session's original target was to compute a twisted trace $Z = \text{Tr}(DT^N)$ of a cylindric transfer matrix, D a diagonal twist, and ask whether the grading it induces on cylindric tableaux is Warnaar's statistic. Theorem 5 answers this without building the model, and for a reason that is *not* the reason I expected.

Corollary 8. *Suppose a vertex model has configurations in bijection with the $(2k, w)$ -cylindric tableaux, with Boltzmann weight x^T , and let D be any operator acting on the winding sector of a configuration by a scalar $t^{\varphi(\cdot)}$. Then $\text{Tr}(DT^N)$ has the form $\sum_T \varepsilon(T) t^{\text{stat}(T)} x^T$ and is therefore subject to Lemma 4(2). At $(k, \ell, n) = (1, 0, 4)$ it cannot equal $(x_1 \cdots x_n)^k \mathcal{G}$.*

(k, ℓ)	n	monomials	(S) viol.	$(S^\pm)$ viol.	worst witness $c_\alpha(t)$
(1, 0)	3	14	0	0	—
(1, 0)	4	41	1	1	$2 - 3t + t^3$; $A=B=2$, TOT=4
(1, 0)	5	122	10	10	$2 - 3t + t^3$; TOT=4
(1, 0)	6	365	61	61	$5 - 9t + 5t^3 - t^6$; TOT=8
(1, 1)	5	122	0	0	—
(1, 1)	6	365	1	0	$5 - 5t^2 + t^4$; $A=5, B=4$, TOT=18
(1, 1)	7	1094	14	0	$5 - 5t^2 + t^4$
(1, 1)	8	3281	113	0	$14 - 20t^2 + 7t^4$; $A=14, B=13$
(1, 2)	7	1094	0	0	—
(1, 2)	8	3281	1	0	$14 - 7t^3 + t^5$; $A=14, B=6$, TOT=68
(1, 2)	9	9842	18	0	$14 - 7t^3 + t^5$
(1, 3)	10	29525	1	0	$42 - 9t^4 + t^6$; $A=42, B=8$, TOT=250
(2, 0)	4	313	145	145	$10 - 16t + 11t^2 - 4t^3$; TOT=11
(2, 0)	5	1563	1051	1051	$27 - 60t + 55t^2 - 20t^3 - 5t^4 + 4t^5$; TOT=21
(2, 1)	6	7813	1817	0	$82 - 90t^2 + 46t^3 - 6t^5$; $A=76, B=44$

Table 1: Exhaustive per-monomial test of Lemma 4. “(S) viol.” counts contents α admitting no stat with HKKO’s signs; “ $(S^\pm)$ viol.” counts those admitting no signed monomial statistic at all. For every monomial of every row the identity $c_\alpha(1) = A_\alpha - B_\alpha$ of Lemma 1 was asserted, and holds. The rows with $n = 2\ell + 4$ and worst witness at $\alpha = (1^n)$ — (1, 0)/4, (1, 1)/6, (1, 2)/8, (1, 3)/10 — are the instances of Theorem 5, whose predicted witnesses $2 - 3t + t^3$, $5 - 5t^2 + t^4$, $14 - 7t^3 + t^5$, $42 - 9t^4 + t^6$ and counts $(C_M, 2M - 2) = (2, 2), (5, 4), (14, 6), (42, 8)$ all appear. Note the dichotomy: at $\ell = 0$ the two violation columns coincide; at $\ell \geq 1$ the second is empty.

That is: the twist ansatz dies, but not because the twist is *diagonal*, and not because its exponent is linear while Warnaar’s is quadratic. It dies on *cardinality*. A twist — diagonal or not, linear or quadratic in the winding — contributes one monomial in t per configuration, and Warnaar’s determinant needs more than one monomial’s worth per configuration. The discrepancy named by the brief turns out to be upstream of the integrable structure entirely.

The secondary, weaker observation remains true and is worth recording, because it says where the quadratic exponent comes from.

Proposition 9. *Let $V = \bigoplus_{a \in L} V_a$ be graded by an abelian group L , let T be any operator on V , and let D act on V_a by $t^{\langle \varphi, a \rangle}$ for a homomorphism $\varphi: L \rightarrow \mathbb{Z}$. Then $\text{Tr}(D T^N) = \sum_{a \in L} t^{\langle \varphi, a \rangle} \text{Tr}_{V_a}(\Pi_a T^N)$, so the exponent is additive in the sector label. Warnaar’s exponent $Q(y) = M \sum_i y_i^2 - \sum_i j y_i$ is not: $Q(2e_1) - 2Q(e_1) = 2M \neq 0$. Hence if the lattice variable y of (1) is identified with the winding sector of a cylindric transfer matrix — the only natural identification, y being the translation part of the affine Weyl group acting on the cylinder — then no diagonal twist reproduces it. A positive-definite quadratic form on the winding lattice is the signature of a level- M theta weight, i.e. of an energy grading t^{L_0} with $L_0 \supset \frac{1}{4M} J_0^2$, not of a flavour twist t^{J_0} .*

I flag the hypothesis explicitly: Proposition 9 is conditional on identifying y with the winding sector, which I have not proved. It is not needed for Corollary 8, which is unconditional.

6 What the statistic must be instead

Theorem 5 says something sharper than “no”.

At $\ell \geq 1$ the strict reading of Warnaar’s question — insert $t^{\text{stat}(T)}$ into the display he quotes — is false, but only just: the deficit is exactly one tableau, at every ℓ . And for $\ell \geq 1$ the re-signed form survives the capacity test with room to spare, the room coming *entirely* from the cylindric shapes $\lambda \in \text{Par}(2k, w)$ with $c_{2k, w}^-(\lambda) = 0$ — at $(1, 1, 6)$, the shape $(4, 2)$, which contributes 9 of the 18 tableaux. So for $\ell \geq 1$ a monomial statistic is not excluded, provided one is willing to recruit shapes that HKKO’s theorem does not mention and to choose signs that are not c^- .

At $\ell = 0$ that escape is closed: there are no intermediate shapes in even degree. So any answer *uniform in ℓ* — and Warnaar’s question is posed uniformly, ℓ being a free nonnegative integer — must abandon the monomial form. It must attach to each cylindric tableau a Laurent *polynomial* $\pi_T(t) \in \mathbb{Z}[t^{\pm 1}]$, not a single power.

This is not an exotic requirement. It is the shape of the classical answer one level down. Macdonald [2, III, (5.8’), (5.9’), (5.11’), pp. 229–230] gives

$$P_{\lambda/\mu}(x; t) = \sum_T \psi_T(t) x^T, \quad \psi_T(t) = \prod_{i \geq 1} \psi_{\lambda^{(i)}/\lambda^{(i-1)}}(t), \quad \psi_{\lambda/\mu}(t) = \prod_{j \in J} (1 - t^{m_j(\mu)}),$$

summed over tableaux of shape λ/μ — a product of factors $1 - t^m$ per tableau, a polynomial, not a power of t . At $\ell = 0$ Warnaar’s determinant *is* (conjecturally, by his (Eq_PCn), which I verified at $(n, k) = (3, 1), (4, 1), (3, 2)$) a sum of Hall–Littlewood polynomials $\sum_{\lambda \text{ even}, \lambda_1 \leq 2k} P_\lambda(x; t)$. So the t in the problem is a Hall–Littlewood bulk parameter, which deforms a tableau count by a product of $(1 - t^m)$ ’s; it is not a boundary fugacity, which would deform it by a power. The quadratic Gaussian $t^{My_i^2 - jy_i}$ on the Weyl-sum side of the identity is what such a bulk weight *sums to* after the affine folding, not what it *is* on the tableau side.

7 Gaps and what is not claimed

1. Theorem 5 is a complete proof of a negative statement, uniform in ℓ , with every number derived in the proof; Theorem 7 is a single verified parameter point. They rest on exactly two external inputs: HKKO Theorem 3.3 (eq. (3)) and the definition (1). They do *not* rest on Warnaar’s Conjecture (Eq_PCn), which is used only as an independent check on my transcription.
2. I have *not* shown that $(S^\pm)$ holds for any $\ell \geq 1$. The capacity test is necessary, not sufficient; failing to violate it is not evidence of existence. Table 1 records a non-violation, which is an absence of evidence.
3. I have *not* constructed the polynomial weight $\pi_T(t)$ that §6 argues must exist. Identifying it is the real form of Warnaar’s question and is open.
4. Proposition 9 is conditional, as flagged there.
5. Trap 5 of the session brief (Orevkov–Orevkov’s $n \leq 14$ degeneracy for multiplicative versus additive polytopes) is recorded as **not applicable**: nothing here is a honeycomb or polytope computation.

References

- [1] J. Huh, J. S. Kim, C. Krattenthaler, S. Okada, *Bounded Littlewood identities for cylindric Schur functions*, arXiv:2301.13117. Theorem 3.3 and §2 as cited above.
- [2] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press. Chapter III, §5, pp. 229–230.
- [3] S. O. Warnaar, arXiv:2511.17034. §6 (Open problems), and Conjecture (Eq_PCn).